22. ROS2 Gazebo Simulation Platform

1. Introduction to Gazebo

Gazebo is the most widely used **3D physics simulation platform** within the ROS ecosystem. It supports physics engines and is capable of high-quality graphical rendering. Beyond merely simulating robots and their surrounding environments, it allows for the incorporation of physical properties such as friction and elasticity coefficients.

For instance, if we were developing a Mars rover, we could simulate the Martian surface environment directly within Gazebo. Similarly, when working with drones—where factors like limited battery life and flight restrictions often prevent frequent experimentation with physical hardware—it is highly practical to utilize Gazebo for initial simulations. Once the algorithms have reached a sufficiently advanced stage of development, they can then be deployed and executed on the actual physical robot.

Consequently, simulation platforms like Gazebo serve as invaluable tools for validating robot algorithms, optimizing robot designs, and testing robot applications within specific scenarios, thereby expanding the possibilities available in robot development.

Note: This chapter is intended solely for informational and learning purposes; the Gazebo environment is *not* configured within this tutorial, as the development process here relies directly on debugging with physical hardware.

2. Installation and Execution

The Gazebo Harmonic version corresponds to ROS2 Jazzy. Since Gazebo Harmonic is not included in the standard ROS package repositories, you must first add the Gazebo Harmonic download source.

```
sudo wget https://packages.osrfoundation.org/gazebo.gpg -O
/usr/share/keyrings/pkgs-osrf-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/pkgs-
osrf-archive-keyring.gpg] http://packages.osrfoundation.org/gazebo/ubuntu-stable
noble main" | sudo tee /etc/apt/sources.list.d/gazebo-stable.list > /dev/null

sudo apt update
```

Proceed to install the Gazebo Harmonic-related plugins.

```
sudo apt install -y gz-harmonic
sudo apt install -y ros-jazzy-ros-gz
```

Add the directory path for the `turtlebot3_gazebo` models to your environment variables.

```
echo 'export
GAZEBO_MODEL_PATH=$GAZEBO_MODEL_PATH:/opt/ros/jazzy/share/turtlebot3_gazebo/mode
ls' >> ~/.bashrc
source ~/.bashrc
```

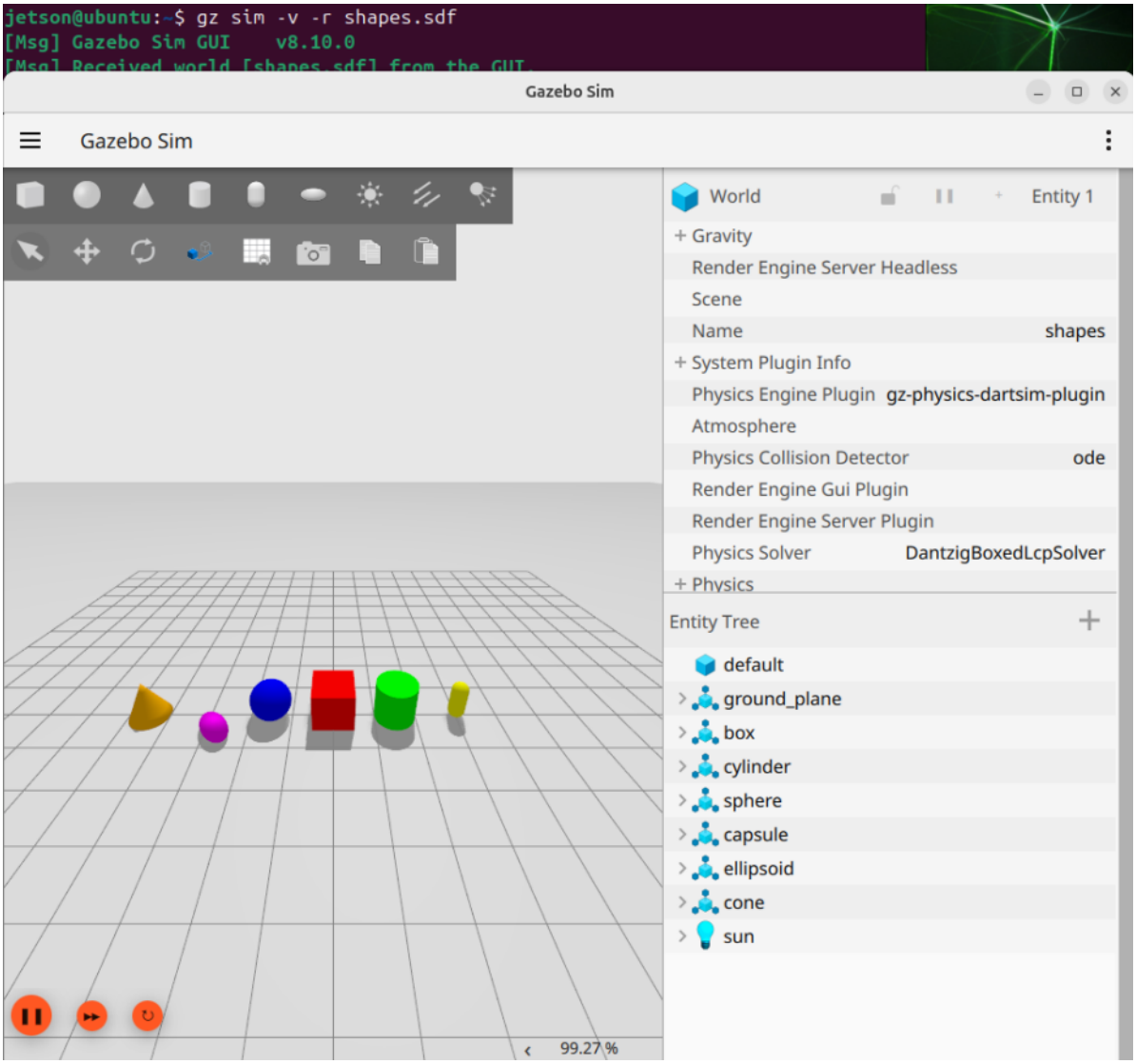
Running Gazebo via the Command Line

Open a terminal and run the `shapes.sdf` test example.

```
gz sim -v -r shapes.sdf
```

Here, the `-v` argument stands for "verbose" mode, which prints detailed logs; `-r` stands for "render," meaning it launches the GUI; and `shapes.sdf` is a test world included with Gazebo Harmonic, located at the path: `/usr/share/gz/gz-sim8/worlds/shapes.sdf`.

Executing this command opens the Gazebo simulation interface, which contains a cone, an ellipsoid, a sphere, a cube, a cylinder, and a capsule. When the simulation runs, the ellipsoid will move slightly.



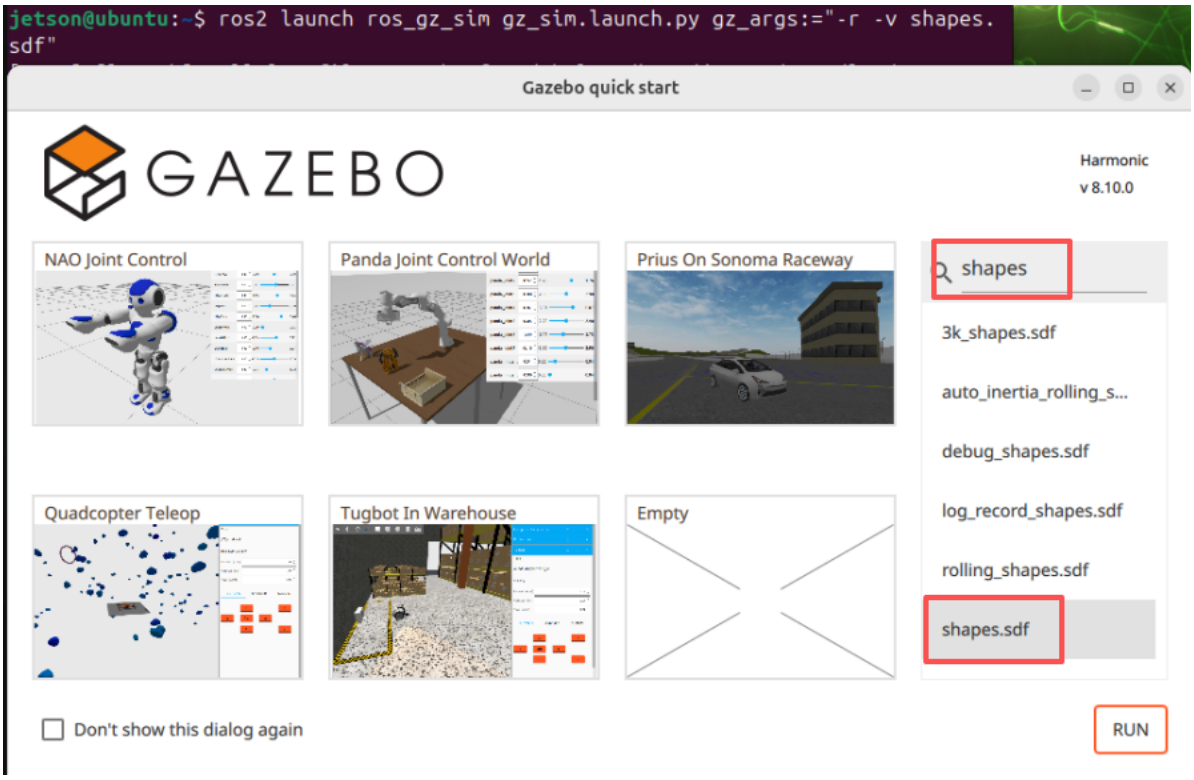
Press `Ctrl+C` in the terminal to exit Gazebo.

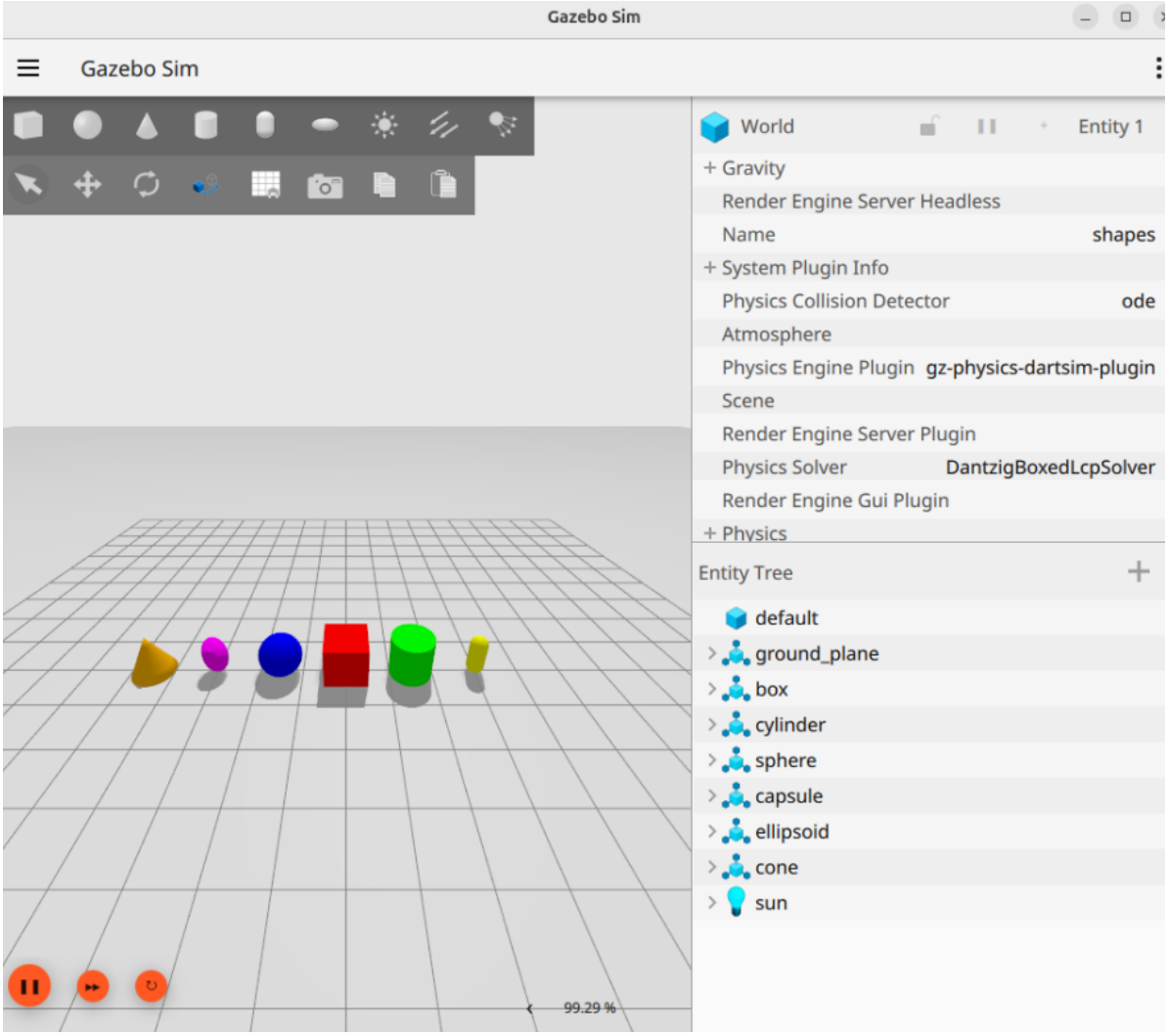
Running Gazebo via ROS

Open a terminal and run the `shapes.sdf` test example.

```
ros2 launch ros_gz_sim gz_sim.launch.py gz_args:="-r -v shapes.sdf"
```

Select the `shapes.sdf` file from the options on the right side, then click `RUN`. You may also select other world files to test.





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